

WESTFIELD REFERENCE RESEARCH · DRAFT REVIEW

Expandable Concertina Cooler

A compact reference deck for foldable, collapsible, and expandable cooler architectures. The goal is to identify practical mechanisms, manufacturable material systems, and risks before concept development.

Design Direction

The working product idea is a cooler that can collapse for storage and expand when the user needs more volume, using a concertina or bellows-like structure. The strongest path appears to be a hybrid construction: controlled folding side structure, segmented insulation, leak-resistant liner strategy, and a carry load path that does not rely on the folding wall.

What the market already validates

Consumers understand the value of compact storage, especially for cars, apartments, RVs, beach gear, grocery runs, sports sidelines, and tailgating. The key opportunity is making collapsibility feel more premium and controlled.

What the concept must solve

A true concertina cooler has to manage insulation gaps, water collection in fold valleys, liner slack, zipper alignment, and handle loads. These are manufacturing problems as much as styling problems.

Reference Set

These examples are not all direct competitors. Some are mechanism references, some are manufacturing references, and some are useful because they show what commodity fold-flat coolers already do.



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COLLAPSED

ORIGAMI SOFT FORM

BUILT Origami Neoprene Tote

Photo reference supplied for the BUILT Origami Neoprene Wine Bottle Tote.

Small neoprene bottle tote that folds from a flat soft form into volume through origami-like geometry.

- Good reference for fold language and compressibility.
- Limited structural and payload relevance for a larger cooler.
- Shows how soft material stretch can hide pattern complexity.



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SEMI-RIGID PANELS

Kelty Folding Cooler

Foldable cooler using fabric, molded PE foam panels, reinforced bottom details, and removable PEVA liner.

- Strong reference for cooler-like structure in a folding format.
- Panel strategy is more manufacturable than continuous insulated bellows.
- Removable liner simplifies cleaning and leak management.



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POP-UP TROLLEY

Enrich Pop-Up Cooler

Supplier example combining collapsible construction, welded leak-proof lining, telescopic handle, and wheels.

- Useful manufacturing reference for welded liner claims.
- Shows how hardware can dominate collapsible cooler architecture.
- Important load-path reminder for larger capacity designs.



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COLLAPSED

CONTROLLED COLLAPSE

CleverMade Tahoe

Collapsible soft cooler with SnapHinge-style side structure, metal rim, baseboard, and insulated liner.

- Clean benchmark for controlled folding and fast setup.
- Collapsed height around 2.5 in is a strong storage story.
- Rim and baseboard help the soft cooler feel intentional.



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COLLAPSED

HARD COOLER PROOF

Coleman Snap 'N Go

New collapsible hard-sided cooler family announced for 35, 45, and 55 qt formats.

- Market proof that collapsible coolers are moving into mainstream hard-cooler territory.
- Claims collapse to one-third size and quick open/collapse action.
- Useful benchmark for consumer expectations around performance.



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COLLAPSED

MASS-MARKET SOFT COOLER

Coleman 54-Can Collapsible

Large fold-flat soft cooler with leak-proof antimicrobial liner, shoulder strap, zipper, and pocketing.

- Useful commodity price and feature benchmark.
- Simple fold-flat format, not much volume adjustability.
- Good reminder that low-cost expectations are already established.



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COMPRESSED

COMPRESSION CONTROL

OGIO Alpha Soft Cooler

Fold-flat 24-can cooler with welded PEVA liner, tote handles, shoulder strap, and top compression strap.

- Compression strap is relevant for locking a smaller volume mode.
- Welded liner language supports the premium soft-cooler direction.
- Useful reference for a cleaner, less commodity visual style.



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ROLLED

ROLL-TOP SOFT BODY

IceMule

Waterproof roll-top soft cooler family that collapses or rolls down when empty.

- Strong reference for leak resistance and portability.
- Roll-top architecture avoids zipper-lid alignment problems.
- Less useful if the target product needs boxy access.



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FLAT STORAGE

FLEXIBLE INSULATION

Nutshell 17L CocoFiber

Foldable cooler using coconut husk fiber insulation and a sustainability story.

- Interesting alternative to foam for repeated fold zones.
- Potentially relevant if rigid foam panels make concertina motion awkward.
- Good sustainability cue if the concept needs a material story.



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CLOSED

COLD PANEL IDEA

FlexiFreeze Flat Cold Cooler

Fold-flat cooler with freezable ice cells integrated into the side structure.

- Suggests turning fold panels into cold-assist panels.
- Could reduce reliance on loose ice in some use cases.
- More relevant as a feature concept than a structure reference.



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FOLDED REF.

SUPPLIER BENCHMARK

64-Can Semi-Rigid Supplier Example

Factory-style semi-rigid folding cooler reference with 600D polyester, heat-welded liner, and stated load capacity.

- Useful for manufacturing language and common material stacks.
- Shows typical supplier claims around leak-proof liners.
- Good reminder to validate real weld quality and load testing.



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COMMODITY MARKET

OPUX / Generic Collapsible Bags

Large soft-sided collapsible cooler bags sold through mass and business retail channels.

- Useful for price, size, feature, and visual baseline.
- Usually not strong mechanism references.
- Important to avoid designing something that feels like a generic bag with folds.

Mechanism Matrix

The references point to several viable families. Concertina should borrow from more than one rather than copying a single product.

Family	Examples	Usefulness for concept
Soft origami fold	BUILT Origami Tote	Useful visual and pattern language, but weak structural precedent.
Structured folding frame	CleverMade Tahoe	Strongest soft-cooler collapse mechanism reference.
Semi-rigid panel system	Kelty Folding Cooler, supplier examples	Most realistic route for insulation and shape control.
Collapsible hard cooler	Coleman Snap 'N Go	Market proof and performance benchmark, though likely different tooling.
Roll/pack soft body	IceMule	Good for waterproofness and portability; less box-like.
Flexible insulation or cold panels	Nutshell, FlexiFreeze	Potential solution for fold zones and cooling performance.

Best Ideas To Borrow

Controlled fold structure

Borrow the idea of snap hinges, rims, baseboards, and defined fold paths so collapse feels deliberate instead of floppy.

Segmented insulation

Use foam or flexible insulation in bays rather than expecting a thick continuous insulated wall to concertina cleanly.

Independent load path

Route handles, shoulder straps, and trolley loads through reinforced straps, rails, or side columns instead of through the bellows wall.

Volume lock positions

Use straps, clips, zippers, snaps, or toggles to hold low, medium, and full-volume states without the cooler ballooning.

Liner strategy

Consider a removable welded bucket liner or slack-managed inner liner to keep the leak zone simpler than the outer folding body.

Base stays rigid

Keep the bottom more rigid and durable than the walls so the cooler can be loaded, dragged, or set down without collapsing unpredictably.

Manufacturing Risks

Concertina-specific issues

- Fold valleys can collect water, dirt, and ice melt.
- Repeated folding can crush foam and create thermal bridges.
- Sharp pleats are bad for welded liners and coated materials.
- Expanded height changes can disturb lid and zipper alignment.

Construction issues

- Welded seam paths need access, pressure, generous radii, and compatible material stacks.
- Too many layers at handle or zipper zones can cause cold welds, burn-through, or cosmetic defects.
- External pockets and heavy hardware add leakage and stress complications.
- Any waterproof claim needs zipper, liner, seam, and QA support.

Recommended concept path

Develop a hybrid concertina cooler: rigid or semi-rigid base, controlled sidewall fold structure, segmented insulation panels, welded removable or slack-managed liner, compression/locking straps for volume settings, and carry loads routed around the folding wall.

Source Links

- [BUILT Origami Neoprene Bottle Tote](#)
- [Kelty Folding Cooler](#)
- [Enrich Bags Pop-Up Cooler](#)
- [CleverMade Tahoe Collapsible Cooler](#)
- [Coleman Snap 'N Go Collapsible Hard Cooler](#)
- [Coleman 54-Can Collapsible Soft-Sided Cooler](#)
- [OGIO Alpha Soft Cooler](#)
- [IceMule Soft Coolers](#)
- [Nutshell 17 L CocoFiber Foldable Cooler](#)
- [FlexiFreeze Refreezable Soft Sided Flat Cold Cooler](#)
- [64-Can Semi-Rigid Folding Cooler Supplier Example](#)
- [OPUX / Generic Collapsible Soft Cooler Example](#)